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Field Development Plan Optimization Under Uncertainty Using Physics-Informed Neural Networks and Generative AI: A Real-Field Case Study in North America

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Abstract

Field Development Plan (FDP) optimization is a notoriously challenging, high-dimensional problem, often involving an immense number of variables and complex operational constraints. Traditional optimization approaches, which rely on iterative reservoir simulations, are computationally expensive and frequently yield sub-optimal solutions. This paper introduces a novel, integrated workflow for FDP optimization under uncertainty, harnessing the capabilities of Generative AI (Gen AI) and advanced physical surrogate models to address these challenges.

The proposed methodology comprises three core components. First, a Generative AI model is employed to recommend realistic well locations, rigorously accounting for all operational constraints and practical considerations. Second, physical surrogate models are developed to emulate the behavior of high-fidelity reservoir simulators. These models are trained using physical reservoir flow concepts, enabling them to predict production rates with remarkable speed—delivering results in milliseconds—while preserving the same accuracy as a high-fidelity simulator. Third, a metaheuristic optimizer is integrated with the physical surrogates to efficiently navigate the high-dimensional optimization landscape and identify the optimal FDP configuration.

This approach overcomes key limitations of conventional deep learning methods, which often require extensive datasets and may fail to honor physical laws. By embedding physical constraints directly into the surrogate model architecture, the physical surrogates ensure both speed and accuracy, bridging the gap between physics-based simulation and data-driven modeling. The Gen AI component is among the first to generate realistic well trajectories that comply with operational restrictions, uncovering new opportunities that traditional workflows might overlook. Furthermore, the use of metaheuristic optimization enables the efficient solution of high-dimensional problems within practical timeframes.

Applied to a real-field complex North America reservoir which comprised 10 completed wells distributed across multiple zones within four distinct reservoir units, the workflow achieved a ~24% increase in expected net present value (NPV) and improved mean oil production across uncertainty scenarios, while maintaining low surrogate prediction error (~3%) even in extrapolation. Alternative uncertainty weightings

revealed distinct optimal well placement patterns with only modest NPV reductions, demonstrating robustness and adaptability.

Introduction

The global decline in new hydrocarbon discoveries has heightened the importance of maximizing recovery from existing reservoirs. Among the various techniques employed to enhance oil production, water injection remains one of the most widely adopted methods for secondary recovery. By maintaining reservoir pressure and displacing hydrocarbons towards production wells, water injection has proven to be a cost-effective and efficient solution, particularly in conventional fields. However, in these contexts, the selection and optimization of well location and trajectory for water injection strategies become exceptionally challenging. Decisions surrounding the optimum field development plans are extremely delicate due to the high number of variables and the complexity of the phenomena involved, such as the intricate interactions between rock and fluid, reservoir heterogeneity, and dynamic changes in reservoir conditions over time. As a result, advanced optimization and modelling techniques are essential to ensure that water injection schemes are both technically sound and economically viable. This paper is motivated by the need for a reliable methodology to determine the optimum field development plan, specifically optimizing the well locations.

Optimizing the placement of water injection FDP in oil reservoirs is an extremely complex and computationally demanding task for reservoir engineers. The challenge arises from the sheer number of variables—such as well locations, injection rates, and scheduling—which can result in more than 10^{100} possible field development scenarios (Isebor et al., 2014). Each scenario typically requires a full reservoir simulation, consuming hours or even days per run, making exhaustive manual evaluation infeasible. As a result, engineers often rely on manual workflows, using quality maps and engineering judgment to select and analyze a limited number of scenarios within project deadlines. This manual process is further complicated by the non-linear and multi-modal nature of objective functions, such as maximizing net present value (NPV) or oil recovery (OR), which makes it difficult to guarantee optimal solutions (Emerick & Reynolds, 2009).

Manual optimization is inherently suboptimal because it explores only a small fraction of the possible solution space and is susceptible to cognitive bias in scenario selection (Guyaguler, 2002). Studies have shown that while traditional methods like quality maps and iterative manual adjustments are convenient, they often fail to account for the dynamic behavior of reservoir fluids over time and may not ensure long-term profitability (Aliyev et al., 2022). For that reason, well location optimization is highly challenging to solve manually, and computational/mathematical techniques are necessary to address this problem.

Decision-making for water injection production strategy has been addressed formally from a mathematical simulation-based optimization perspective (Torrado et al., 2015). Torrado et al. (2015) introduce an efficient technique for optimizing water injection strategies under operational constraints. While the method enables rapid evaluation through advanced mathematical optimization and reservoir simulation, the authors acknowledge a key limitation: the approach explores a suboptimal solution space to achieve computational speed. This trade-off prioritizes fast performance evaluation over identifying the absolute optimal solution, making it particularly valuable for scenarios requiring rapid decision-making (e.g., real-time adjustments or large-scale screening) where near-optimal solutions are acceptable.

Early work (Emerick et al. 2009) demonstrated genetic algorithms for injector placement in China's Pubei field using advanced mixed-integer nonlinear programming (MINLP) for discrete well placement decisions. These foundational methods established computational frameworks but faced limitations in handling high-dimensional search spaces and geological uncertainty.

Methodological advancements in reservoir optimization have been significantly driven by the integration of hybrid optimization techniques. These approaches enabled more efficient and robust solutions to complex field development planning challenges. For example, (Isebor et al. 2014) introduced a generalized

framework for the joint optimization of well placement and control strategies, demonstrating improved performance over traditional sequential methods. Similarly, (Forouzanfar et al, 2010) proposed a two-stage optimization approach that effectively balances exploration and exploitation phases, enhancing the convergence and quality of solutions in high-dimensional search spaces. The adoption of such hybrid methods substantially reduced computational requirements and improved the feasibility of large-scale reservoir optimization problems.

Further developments in this area leveraged advanced metaheuristics and surrogate modeling techniques to accelerate optimization workflows. For instance, (Chen et al. 2009) utilized evolutionary algorithms in conjunction with response surface methodologies to reduce the number of required reservoir simulations, achieving significant reductions in computational time. Additionally, (Bangerth et al. 2006) demonstrated the utility of adjoint-based optimization for well placement, providing gradient information that further streamlines the search process. Collectively, these methodological breakthroughs transformed field development planning, enabling practitioners to address increasingly complex reservoir scenarios with greater efficiency and accuracy.

In recent years, the integration of artificial intelligence (AI) and machine learning (ML) techniques has become central to accelerating reservoir simulation and optimization workflows. Notably, convolutional neural network surrogates have enabled simulation runtimes to be reduced by up to 80% (Zhang et al., 2022). Cloud-based distributed optimization frameworks now support the evaluation of over one million scenarios. Additionally, ML-driven approaches have made it possible to incorporate CO₂ emission constraints into joint well-control optimization (Zheng et al., 2024). The development of graph network surrogates has resulted in evaluations for storage reservoirs that are up to 120 times faster (Tang, 2024), and Neural Operators have been used to generate maps of pressure and saturation (Badawi et al., 2025). Collectively, these advances highlight the transformative impact of AI and ML on computational efficiency and multi-objective optimization in reservoir engineering.

It should be stressed that, in general, all optimization algorithms require a first stage where the search space needs to be explored globally. After this first exploration stage, the solution found is often incrementally improved until some optimization stopping criterion is satisfied. The selection of the algorithm has a strong dependency on the type of problem to solve, the size of the search space (or in other words, the number of variables to optimize), its robustness to introduce constraints of different types, and the search/exploration capabilities.

However, all these optimization techniques present several common drawbacks:

- Optimizers depend on a highly computationally expensive finite volume reservoir simulator, where a simulation can take hours or days. Additionally, optimizers need to call hundreds of thousands or millions of times to the reservoir simulator in order to reach a local/global optimum. The combination of both issues implies that the majority of the optimizers cannot be used for well placement locations or if they are, the solution they reach is expected to be very sub-optimal.
- One of the main challenges for well placement optimization is to create an algorithm that generates well-locations that are geologically plausible, in other words, the optimization algorithms provide solutions that drilling engineering, and geoscientists consider realistic to implement directly in the field.
- One of the main challenges of current optimization techniques is managing realistic operational constraints. From the mathematical point of view, these non-linear constraints are complicated to manage in the optimizer and affect the convergence of the problem.
- Machine learning techniques have primarily been tested in small-scale reservoir models that lack geological realism or integration of complex operational constraints, such as inter-well distances and azimuth angles. Furthermore, these approaches typically fail to incorporate physical principles into deep neural network architectures for characterizing porous media flow dynamics.

In this paper, we propose a new optimization workflow based on artificial intelligence to address all of these drawbacks, as described in [patent WO 2025/081114 A2 \(PCT\)](#). Our workflow tackles each limitation individually. First, we implement an AI-driven optimization algorithm that ensures all wells comply with operational constraints. Second, we introduce a novel surrogate modeling technology that enables simulations to run orders of magnitude faster than traditional reservoir simulation methods, without sacrificing accuracy. Finally, we integrate the surrogate model with generative AI for well trajectory generation within an optimization workflow to maximize a dynamic performance variable (e.g., Net-Present-Value), thereby determining the optimal production strategy.

This paper is organized as follows: first, we present the methodology; second, we test our workflow on a real reservoir in North America; and finally, we provide conclusions and outline future work.

Methodology

Our methodology is illustrated in [Figure 1](#). As depicted in the workflow, the process comprises three primary components: the artificial intelligence (AI) module responsible for generating well locations, the surrogate model, and the optimizer

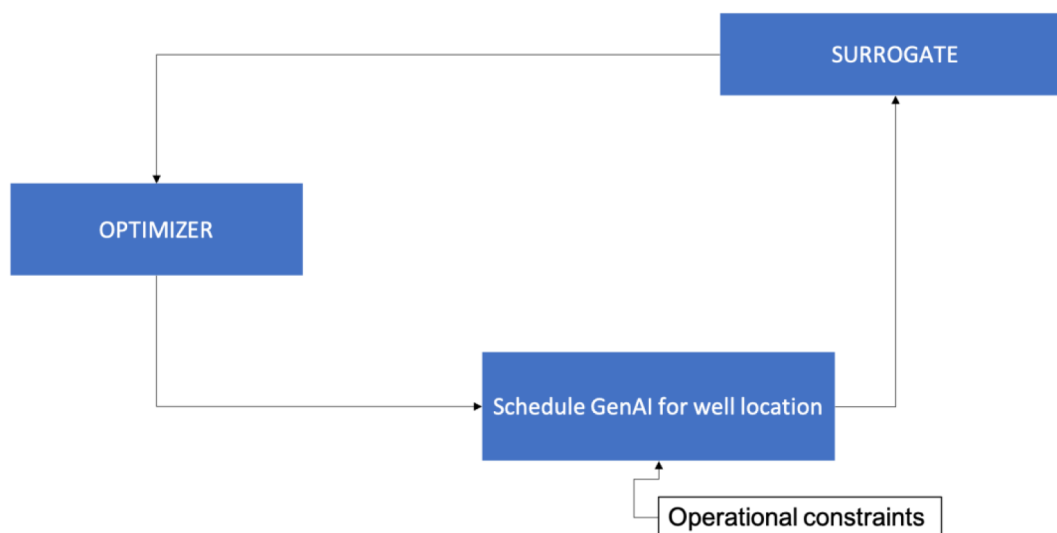


Figure 1—General workflow of the methodology for well placement using artificial intelligence.

Generative well locations: Well optimization algorithms need to assess potential well locations. Some of these locations are unlikely to yield favorable outcomes due to reservoir characteristics or a violation of operational limitations, such as dogleg severity. In other words, we can prune the original action space to focus on solutions that are both geologically promising and operationally feasible. As described in [Patent WO 2025/081114 A2 \(PCT\)](#), we propose a generative AI-based optimization algorithm that generates a set of well trajectories, all of which satisfy both reservoir and operational constraints within our workflow. This is done given a set of variables y that parameterize and define the well trajectory, and the AI algorithm guarantees that we are respecting the operational constraints

Optimizer and surrogate model: our objective is to determine the final optimal trajectory that maximizes a dynamic property such as NPV, employing optimization algorithms like evolutionary meta-heuristic techniques. We formally approach this challenge from a simulation-based mathematical optimization perspective; however, this method is constrained by the significant computational demands of commercial reservoir simulators, which restrict the number of scenarios that can be practically evaluated. To address this limitation, we incorporate surrogate physics modeling as a key component of our workflow, enabling efficient approximation of reservoir behavior and broader scenario exploration. Additionally, we propose

a modification to evolutionary algorithms—specifically, the Covariance Matrix Adaptation Evolution Strategy (CMA-ES)—to ensure compliance with complex operational nonlinear constraints throughout the optimization process.

This workflow face increased challenges when incorporating geological uncertainty. Reservoir simulations depend on input from geological models—such as porosity and permeability maps—generated through geostatistics using sparse well data. This limited data leads to multiple equiprobable geological realizations that can represent the reservoir's behavior. Mathematically, uncertainty in well placement optimization is addressed by calculating the expected value of an objective function (e.g., NPV). This involves simulating every production strategy across all possible realizations and averaging the results—a computationally intensive process, especially for complex reservoirs. Due to the speed and accuracy of our surrogate model, we are not constrained by the limitations of traditional high-fidelity simulators, allowing us to simulate the expected value without adding significant computational cost.

In the following section, we will focus on optimization and surrogate modelling. Please note that a detailed description of the methodology can be found in [patent WO 2025/081114 A2 \(PCT\)](#).

Optimizer and surrogate model

The objective of this workflow is to obtain a set of producers and injectors $p, q \in W$ such that we maximize a dynamic variable (e.g. NPV or Oil Production (Q_o)). We can formally address this problem from a mathematical simulation-based optimization perspective, where NP is the maximum number of producer wells to drill and NI is the maximum number of injectors to drill, and p and q must respect several operational constraints such as maximum and minimum distances between producers, maximum and minimum distance between producer-injectors, and maximum and minimum distance between injectors.

Calculating a dynamic variable such as Q_o presents a primary challenge in this problem. Traditionally, Q_o is computed using computationally expensive reservoir simulators. This creates a significant bottleneck, as optimization techniques typically require hundreds of thousands of simulator runs – a prohibitively costly approach.

To address this limitation, we propose a novel deep learning-based physics-inspired surrogate modeling approach. Our surrogate achieves simulation speeds of 0.0001 seconds per run while maintaining comparable accuracy for well rates. A description of this surrogate model is available in ([Torrado et al. 2022](#)). The second critical component of the closed-loop optimization workflow is the optimizer. In this study, we implement the Covariance Matrix Adaptation Evolution Strategy (CMA-ES).

Finally, combining both generative AI and surrogate optimization workflows, we are able to optimize under uncertainty the number of producer and injector wells, well location, and well trajectory, maximizing a dynamic variable (e.g. NPV) and respecting the operational constraints, which allows us to apply our solution in the reservoir.

Results

To test the workflow presented in the paper, we proposed to validate it on a complex North American reservoir. This field comprises 10 completed wells distributed across multiple zones within four distinct reservoir units. Development planning is complicated by significant geological uncertainty, which is characterized using LOW, MID, and HIGH cases to represent a range of facies distributions and connectivity scenarios.

Operational constraints further increase project complexity. Strict limitations on well inclination, azimuth, dogleg severity, and inter-well spacing must be enforced across vertical, high-angle, and horizontal drilling operations. The reservoir is intersected by multiple faults, and all new wellbores must comply with minimum offset requirements from these faults, as well as specified minimum and maximum distances between producer-producer and producer-injector pairs.

These technical constraints, combined with the requirement to integrate existing infrastructure and maintain reservoir connectivity, create a highly challenging context for well placement optimization and development planning. Drilling activities for new wells were scheduled to commence as early as October 2018, necessitating robust and adaptive planning to accommodate both geological uncertainty and operational limitations.

In this paper, we specifically focus on NPV optimization of well locations and trajectories for a fixed configuration of 1 producer and 1 injector, followed by our initial assessment of two cases: 1 producer 1 injector and 2 producer and 2 injectors for the MID case. The results section is structured to analyze the outcomes of each component of the workflow: surrogate modeling, and the overall optimization process.

Surrogate Performance

To increase the accuracy of the surrogate, we developed a separate surrogate for each uncertainty case, i.e. LOW, MID and HIGH. For training the surrogate model, we use a dataset comprising 1000, 2500 and 1500 simulations generated with a high-fidelity simulator for LOW, MID and HIGH cases respectively. Additionally, we employ a separate set of 200 randomly selected realizations for a testing dataset to evaluate the surrogate's performance through blind testing. To ensure an unbiased assessment of the surrogate's quality, our validation focuses on this testing dataset.

Table 1 summarizes the three Mean Absolute Percentage Error (MAPE = $\sum_{i=1}^N (y_i - \hat{y})_i / y_i$) ranges of the surrogate model for the field variable FOPT for LOW, MID and HIGH cases. The FOPT model achieves a 50 percentile (P50), and 90 percentile (P90), error of 0.3%, 1.0% for LOW compared with 1.7 %, 2.5% for MID and 1.0 %, 5.0%, for HIGH cases respectively. Because oil has a substantially higher price per barrel than the unit cost of water injection or Water or Gas production, we decided only to consider the FOPT term in NPV calculation during optimization with surrogate and later on to include Water and Gas terms in the actual NPV from the simulator.

Table 1—Summary of MAPE statistics for each surrogate model (all values in %)

Reported Error	FOPT (LOW)	FOPT (MID)	FOPT (HIGH)
25 Percentile	0.1	0.3	0.4
50 Percentile	0.3	0.7	1.0
75 Percentile	0.4	1.2	2.0
90 Percentile	1.0	2.5	5.0

These results validate the use of a surrogate model for optimization, as it successfully predicts well configurations and trajectories not encountered during training. A key question addressed in the following section is the surrogate's extrapolation capability—specifically, its performance in forecasting oil/water production/injection beyond the training and testing datasets.

Optimization Results

In this section, we present the performance of our optimization workflow under geological uncertainty. Our CMA-ES algorithm is initialized with a population of 5,000 cases, using a crossover probability of 80% and a mutation probability of 20%. Consequently, each iteration of the workflow involves $5,000 \times 3 = 15,000$ simulations or surrogate model evaluations. The initial population was taken from the production strategy (well location) used during the training of the physical surrogate, and we computed some random realizations to reach the 5,000 cases.

The optimal solution identified by the optimizer was re-evaluated using the high-fidelity simulator. We calculated the expected net present value (E(NPV)) across 3 realizations of LOW, MID and HIGH with equal probability, resulting in a value of 24.6 MM USD/STB MM. This is around 24% increase from the base case with E(NPV) of 19.9 MM USD/STB MM. More details are shown in Table 2. In the base case, FOPT ranges from 1.0 in the low scenario up to 1.38 in the high case, with a mean FOPT of 1.185. The optimized strategy slightly increases the low-case FOPT to 1.014 and achieves higher mid and high values, ultimately improving the mean FOPT to 1.29. This improvement is also reflected in the NPV, which rises from 19.9 to 24.6 (in normalized units), though it comes with an additional CAPEX cost of 4.0 units. Also, Figure 2 illustrates how the base and optimized development strategies compare over time in terms of total oil, water, and gas production, as well as water injection, all shown as normalized values. Across all four plots, the optimized case consistently shows higher mean curves than the base scenario. This suggests that the optimized plan leads to more efficient resource recovery and overall production enhancement. The upward shift in the mean lines for oil and gas production indicates improved hydrocarbon extraction, while the higher water production and injection reflect the additional support needed to maintain reservoir pressure and maximize output. These trends clearly demonstrate the benefits of the optimized approach in boosting performance under different uncertainty scenarios.

Additionally, it is important to note that the surrogate model exhibited an error of approximately 3.0% when comparing the best solution identified by the surrogate during the optimization process to the same solution re-evaluated using the high-fidelity simulator. This result demonstrates the surrogate's effectiveness in extrapolation and exploration throughout the optimization process, as it maintains an error level comparable to that observed in our blind test during training.

To understand the effect of uncertainty cases on the NPV and location of wells and trajectories, we repeat the optimization with additional weights (un-equal weights): (1) First case with weights of 0.2, 0.6 and 0.2 for LOW, MID and HIGH case and (2) second case with weight of 0.0, 1.0 and 0.0 for LOW, MID and HIGH cases. Note that the second case is an extreme case where only one realization (MID) is considered. Although we observed a small reduction in NPV (2% for 0.2-0.6-0.2 and around 6.5% for 0.0-1.0-0.0), the regions of interest in the reservoir for the location of producers and injectors are different. For instance, for the case with equal weights, we observed more producers in the center and north of the reservoir while for 0-1-0 they are more in the south. This demonstrates the flexibility of our approach for providing multiple solutions and scenarios with only a minor trade-off in the target values.

Table 2—Normalized (divided by LOW FOPT of base case) FOPT, NPV and CAPEX for Base and optimized cases.

	FOPT (STB MM)			Normalized Mean FOPT (STB/STB)	NPV (Millions/STB MM)	CAPEX (Millions/STB MM)
	LOW	MID	HIGH			
Base (no infill)	1.0	1.18	1.38	1.185	19.9	0.0
FDP Optimization (Equal weight)	1.014	1.27	1.6	1.29	24.6	4.0
FDP Optimization (Un-equal weight) 0.2-0.6-0.2	1.014	1.28	1.61	1.29	24.1	4.0
FDP Optimization (Un-equal weight) 0.0-1.0-0	-	1.27	-	1.27	23.0	4.0

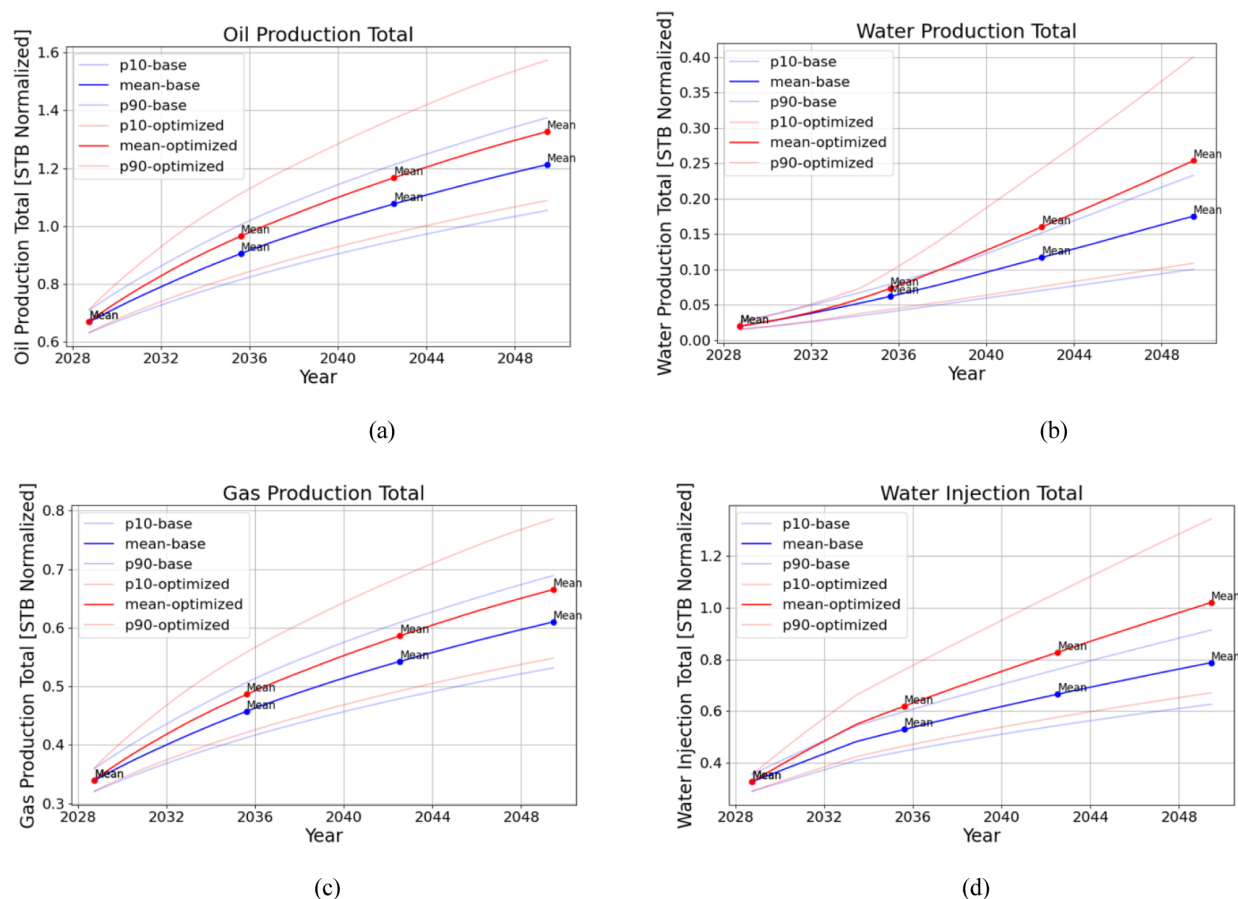


Figure 2—Demonstrates P10, Mean, P90 of base and optimized cases for (a) Oil production total, (b) Water production total, (c) Gas production total and (d) Water Injection total.

Conclusions

In this paper, we present a new methodology for optimizing field development plans using artificial intelligence. Our workflow consists of three main components: an optimizer, generative AI for creating wells that respect operational constraints, and a surrogate physics model to predict oil and water production or injection.

The proposed methodology comprises three core components. First, a Generative AI model is employed to recommend realistic well locations, rigorously accounting for all operational constraints and practical considerations. Second, physical surrogate models are developed to emulate the behavior of high-fidelity reservoir simulators. These models are trained using physical reservoir flow concepts, enabling them to predict production rates with remarkable speed—delivering results in milliseconds—while preserving the same accuracy as a high-fidelity simulator. Third, a metaheuristic optimizer is integrated with the physical surrogates to efficiently navigate the high-dimensional optimization landscape and identify the optimal FDP configuration.

In this paper, we focus on optimizing field development plans for well location/trajectory under uncertainty in a real reservoir in North America comprising 10 completed wells distributed across multiple zones within four distinct reservoir units. Our results show that we can create an accurate surrogate model with an error smaller than 3% for oil/water production/injection. Importantly, this error applies to both interpolation and extrapolation—meaning we can predict oil production rates for field development plans (FDPs) not encountered during training.

The locations of an additional one producer and one injector wells were optimized using over 2,000,000 surrogate evaluations, leading to a ~24% increase in expected NPV compared to the base case. The

optimized plan also achieved higher mean oil production across all uncertainty scenarios. We further investigated the impact of assigning unequal weights to the LOW, MID, and HIGH geological cases during optimization. Weight configurations such as 0.2–0.6–0.2 and the extreme case 0–1–0 resulted in modest NPV reductions of only ~2% and ~6.5%, respectively, while producing distinctly different optimal well placement patterns. These findings highlight the flexibility of the proposed approach—not only in identifying high-value solutions under equal-probability scenarios, but also in tailoring strategies to prioritize specific uncertainty cases without significantly compromising economic performance.

Overall, this integrated Gen AI–surrogate–optimizer workflow provides a computationally efficient and operationally realistic solution for high-dimensional FDP optimization under uncertainty, with demonstrated adaptability to varying economic and geological priorities.

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